



EE 354/CE 361/MATH 310 -
Introduction to Probability
and Statistics
Fall 2025



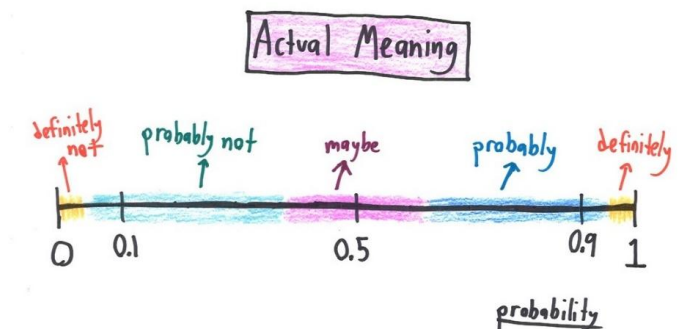


EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

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Week 12 - Lecture No 23 – November 05, 2025



Announcements!

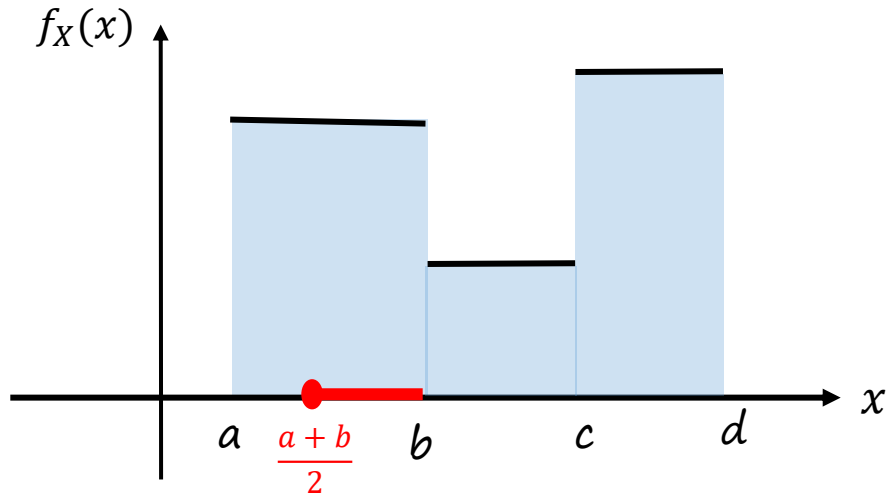
1. All lectures and grades posted.
2. Assignment No 02 uploaded – **due date November 10, 2025** (Assignment 3 to be uploaded by 2nd week of November)
 - A small correction has been made in Question No 13
3. **Quiz No 08 – November 05, 2025 (from Unit 5)**

Agenda for today

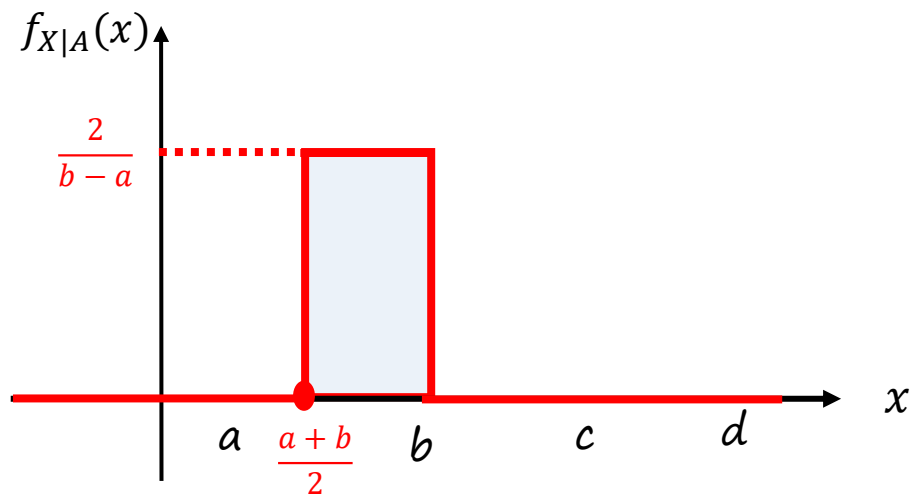
1. Unit 5 – Continuous Random Variables
 - Normal Random Variables – Gaussian Distribution

[illegible]

Example



$$A: \quad \frac{a+b}{2} \leq X \leq b$$



$$\Rightarrow E[X | A] = \frac{1}{2} \frac{a+b}{2} + \frac{1}{2} b$$

$$\Rightarrow = \frac{1}{4} a + \frac{3}{4} b$$

$$\Rightarrow E[X^2 | A] = \int_{\frac{a+b}{2}}^b \frac{2}{b-a} x^2 dx$$

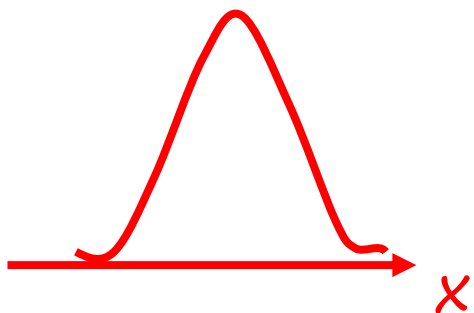
Exercise

Assume $X \sim N(0, 1)$

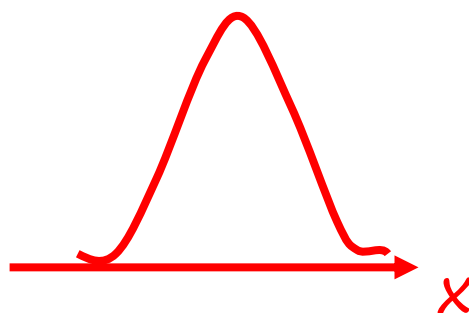
Draw $f_{X|A}(x)$

Where $A =$

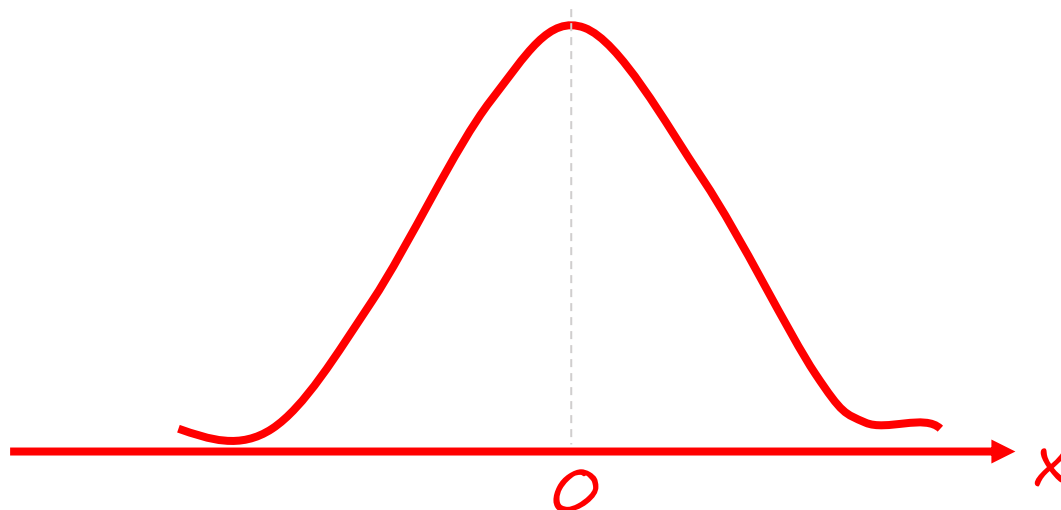
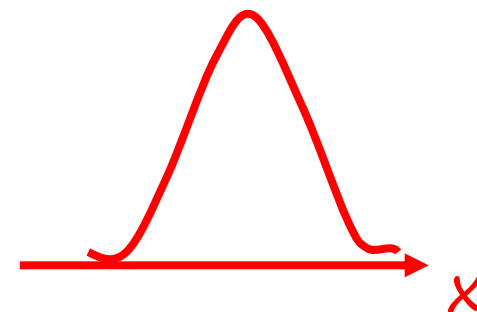
a) $X \in \pm \sigma$



b) $X \in \pm \frac{\sigma}{2}$



c) $X \in \pm \frac{\sigma}{100}$



Memorylessness of the exponential PDF

- Bulb lifetime T : exponential (λ)
- Do you prefer a used or new “exponential light bulb”? *Probabilistically identical!*

$$P(T > x) = e^{-\lambda x}, \text{ for } x \geq 0$$

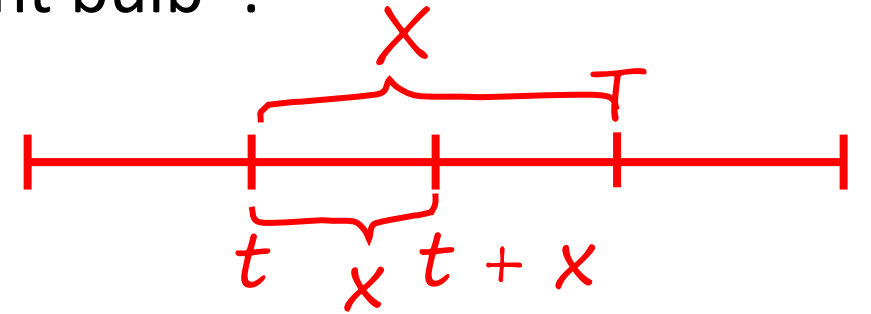
- We are told that $T > t$

- r.v. X : remaining lifetime $= T - t$

$$P(X > x \mid T > t) = e^{-\lambda x}, \text{ for } x \geq 0$$

$$= \frac{P(T - t > x, T > t)}{P(T > t)} = \frac{P(T > t + x, T > t)}{P(T > t)} = \frac{P(T > t + x)}{P(T > t)}$$

$$= \frac{e^{-\lambda(t+x)}}{e^{-\lambda t}} = e^{-\lambda x}$$



Memorylessness of the exponential PDF

$$f_T(x) = \lambda e^{-\lambda x}, \text{ for } x \geq 0$$

$$P(0 \leq T \leq \delta) \approx \cancel{f_T(0)} * \delta = \lambda \delta$$

$$P(t \leq T \leq t + \delta \mid T > t) = P(0 \leq T \leq \delta) \approx \lambda \delta$$

similar to an independent coin flip,

Every δ time steps,

With $P(\text{success}) \approx \lambda \delta$

Can human life be modelled like this?

